



Advanced Linear Algebra: Foundations to Frontiers

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1.3.8 Submultiplicative norms ¶

01.3.8 Submultiplicative norms

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There are a number of properties that we would like for a matrix norm to have (but not all norms do have). Recalling that we would like for a matrix norm to measure by how much a vector is "stretched," it would be good if for a given matrix norm, $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, there are vector norms $\|\cdot\|_\mu : \mathbb{C}^m \rightarrow \mathbb{R}$ and $\|\cdot\|_\nu : \mathbb{C}^n \rightarrow \mathbb{R}$ such that, for arbitrary nonzero $x \in \mathbb{C}^n$, the matrix norm bounds by how much the vector is stretched:

$$\frac{\|Ax\|_\mu}{\|x\|_\nu} \leq \|A\|$$

or, equivalently,

$$\|Ax\|_\mu \leq \|A\| \|x\|_\nu$$

where this second formulation has the benefit that it also holds if $x = 0$. When this relationship between the involved norms holds, the matrix norm is said to be subordinate to the vector norms:

Definition 1.3.8.1. Subordinate matrix norm. A matrix norm $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is said to be subordinate to vector norms $\|\cdot\|_\mu : \mathbb{C}^m \rightarrow \mathbb{R}$ and $\|\cdot\|_\nu : \mathbb{C}^n \rightarrow \mathbb{R}$ if, for all $x \in \mathbb{C}^n$,

$$\|Ax\|_\mu \leq \|A\| \|x\|_\nu.$$

If $\|\cdot\|_\mu$ and $\|\cdot\|_\nu$ are the same norm (but perhaps for different m and n), then $\|\cdot\|$ is said to be subordinate to the given vector norm.

Fortunately, all the norms that we will employ in this course are subordinate matrix norms.

Homework 1.3.8.1. ALWAYS/SOMETIMES/NEVER: The matrix 2-norm is subordinate to the vector 2-norm.

ALWAYS/SOMETIMES/NEVER: The Frobenius norm is subordinate to the vector 2-norm.

► [Answer](#) ► [Solution](#)

Theorem 1.3.8.2. *Induced matrix norms, $\|\cdot\|_{\mu,\nu} : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, are subordinate to the norms, $\|\cdot\|_\mu$ and $\|\cdot\|_\nu$, that induce them.*

Proof.

Corollary 1.3.8.3. *Any matrix p -norm is subordinate to the corresponding vector p -norm.*

Another desirable property that not all norms have is that

$$\|AB\| \leq \|A\|\|B\|.$$

This requires the given norm to be defined for all matrix sizes.

Definition 1.3.8.4. Consistent matrix norm. A matrix norm $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is said to be a consistent matrix norm if it is defined for all m and n , using the same formula for all m and n .

Obviously, this definition is a bit vague. Fortunately, it is pretty clear that all the matrix norms we will use in this course, the Frobenius norm and the p -norms, are all consistently defined for all matrix sizes.

Definition 1.3.8.5. Submultiplicative matrix norm. A consistent matrix norm $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is said to be submultiplicative if it satisfies

$$\|AB\| \leq \|A\|\|B\|.$$

Theorem 1.3.8.6. *Let $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}$ be a vector norm defined for all n . Define the corresponding induced matrix norm as*

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|.$$

Then for any $A \in \mathbb{C}^{m \times k}$ and $B \in \mathbb{C}^{k \times n}$ the inequality $\|AB\| \leq \|A\|\|B\|$ holds.

In other words, induced matrix norms are submultiplicative. To prove this theorem, it helps to first prove a simpler result:

Lemma 1.3.8.7. Let $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}$ be a vector norm defined for all n and let $\|\cdot\| : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ be the matrix norm it induces. Then $\|Ax\| \leq \|A\|\|x\|$.

Proof.

If $x = 0$, the result obviously holds since then $\|Ax\| = 0$ and $\|x\| = 0$. Let $x \neq 0$. Then

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} \geq \frac{\|Ax\|}{\|x\|}.$$

Rearranging this yields $\|Ax\| \leq \|A\|\|x\|$.

We can now prove the theorem:

Proof.

$$\begin{aligned} & \|AB\| \\ &= \text{< definition of induced matrix norm >} \\ & \max_{\|x\|=1} \|ABx\| \\ &= \text{< associativity >} \\ & \max_{\|x\|=1} \|A(Bx)\| \\ &\leq \text{< lemma >} \\ & \max_{\|x\|=1} (\|A\|\|Bx\|) \\ &\leq \text{< lemma >} \\ & \max_{\|x\|=1} (\|A\|\|B\|\|x\|) \\ &= \text{< \|x\| = 1 >} \\ & \|A\|\|B\|. \end{aligned}$$

Homework 1.3.8.2. Show that $\|Ax\|_\mu \leq \|A\|_{\mu,\nu}\|x\|_\nu$ for all x .

▼ Solution

W.l.o.g. assume that $x \neq 0$. (Why? Because if $x = 0$ then obviously $\|Ax\|_\mu \leq \|A\|_{\mu,\nu}\|x\|_\nu$.)

$$\|A\|_{\mu,\nu} = \max_{y \neq 0} \frac{\|Ay\|_\mu}{\|y\|_\nu} \geq \frac{\|Ax\|_\mu}{\|x\|_\nu}.$$

Rearranging this establishes the result.

Homework 1.3.8.3. Show that $\|AB\|_\mu \leq \|A\|_{\mu,\nu} \|B\|_{\nu,\mu}$.

▼ **Solution**

$$\begin{aligned} \|AB\|_\mu &= \text{< definition >} \\ \max_{\|x\|_\mu=1} \|ABx\|_\mu &\leq \text{< last homework >} \\ \max_{\|x\|_\mu=1} \|A\|_{\mu,\nu} \|Bx\|_\nu &= \text{< algebra >} \\ \|A\|_{\mu,\nu} \max_{\|x\|_\mu=1} \|Bx\|_\nu &= \text{< definition >} \\ \|A\|_{\mu,\nu} \|B\|_{\nu,\mu} \end{aligned}$$

Homework 1.3.8.4. Show that the Frobenius norm, $\|\cdot\|_F$, is submultiplicative.

▼ **Solution**

$$\begin{aligned}
& \|AB\|_F^2 \\
&= \text{< partition >} \\
& \left\| \begin{pmatrix} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix} \begin{pmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{pmatrix} \right\|_F^2 \\
&= \text{< partitioned matrix-matrix multiplication >} \\
& \left\| \begin{pmatrix} \tilde{a}_0^T b_0 & \tilde{a}_0^T b_1 & \cdots & \tilde{a}_0^T b_{n-1} \\ \tilde{a}_1^T b_0 & \tilde{a}_1^T b_1 & \cdots & \tilde{a}_1^T b_{n-1} \\ \vdots & \vdots & & \vdots \\ \tilde{a}_{m-1}^T b_0 & \tilde{a}_{m-1}^T b_1 & \cdots & \tilde{a}_{m-1}^T b_{n-1} \end{pmatrix} \right\|_F^2 \\
&= \text{< definition of Frobenius norm >} \\
& \sum_i \sum_j |\tilde{a}_i^T b_j|^2 \\
&= \text{< definition of Hermitian transpose vs transpose >} \\
& \sum_i \sum_j |\tilde{a}_i^H b_j|^2 \\
&\leq \text{< Cauchy-Schwarz inequality >} \\
& \sum_i \sum_j \|\tilde{a}_i\|_2^2 \|b_j\|_2^2 \\
&= \text{< algebra and } \|\bar{x}\|_2 = \|x\|_2 \text{ >} \\
& \left(\sum_i \|\tilde{a}_i\|_2^2 \right) \left(\sum_j \|b_j\|_2^2 \right) \\
&= \text{< previous observations about the Frobenius norm >} \\
& \|A\|_F^2 \|B\|_F^2
\end{aligned}$$

Hence $\|AB\|_F^2 \leq \|A\|_F^2 \|B\|_F^2$. Taking the square root of both sides leaves us with $\|AB\|_F \leq \|A\|_F \|B\|_F$.

This proof brings to the forefront that the notation $\tilde{a}_i^T$ leads to some possible confusion. In this particular situation, it is best to think of $\tilde{a}_i$ as a vector that, when transposed, becomes the row of A indexed with i . In this case, $\tilde{a}_i^T = \tilde{a}_i^H$ and $(\tilde{a}_i^T)^H = \overline{\tilde{a}_i}$ (where, recall, $\bar{x}$ equals the vector with all its entries conjugated). Perhaps it is best to just work through this problem for the case where A and B are real-valued, and not worry too much about the details related to the complex-valued case...

Homework 1.3.8.5. For $A \in \mathbb{C}^{m \times n}$ define

$$\|A\| = \max_{i=0}^{m-1} \max_{j=0}^{n-1} |\alpha_{i,j}|.$$

1. TRUE/FALSE: This is a norm.
2. TRUE/FALSE: This is a consistent norm.

▼ **Answer**

▼ **Solution**

1. This is a norm. You can prove this by checking the three conditions.
2. It is a consistent norm since it is defined for all m and n .

1. TRUE
2. TRUE

Remark 1.3.8.8. The important take-away: The norms we tend to use in this course, the p -norms and the Frobenius norm, are all submultiplicative.

Homework 1.3.8.6. Let $A \in \mathbb{C}^{m \times n}$.

ALWAYS/SOMETIMES/NEVER: There exists a vector

$$x = \begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{n-1} \end{pmatrix} \text{ with } |\chi_i| = 1 \text{ for } i = 0, \dots, n-1$$

such that $\|A\|_\infty = \|Ax\|_\infty$.

▼ [Answer](#) ▼ [Solution](#)

Partition A by rows:

$$A = \begin{pmatrix} \tilde{a}_0^T \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix}.$$

We know that there exists k such that $\|\tilde{a}_k\|_1 = \|A\|_\infty$. Now

$$\begin{aligned} \|\tilde{a}_k\|_1 &= \text{< definition of 1-norm >} \\ |\alpha_{k,0}| + \dots + |\alpha_{k,n-1}| &= \text{< algebra >} \\ \frac{|\alpha_{k,0}|}{\alpha_{k,0}} \alpha_{k,0} + \dots + \frac{|\alpha_{k,n-1}|}{\alpha_{k,n-1}} \alpha_{k,n-1}. \end{aligned}$$

where we take $\frac{|\alpha_{k,j}|}{\alpha_{k,j}} = 1$ whenever $\alpha_{k,j} \neq 0$. Vector

$$x = \begin{pmatrix} \frac{|\alpha_{k,0}|}{\alpha_{k,0}} \\ \vdots \\ \frac{|\alpha_{k,n-1}|}{\alpha_{k,n-1}} \end{pmatrix}$$

has the desired property.

ALWAYS

Now prove it!